

ZHUOCHENG DUAN

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Tel: 0410615608

Date of Birth: 31 July 1997

Nationality: Chinese

 [Personal Website](#)

Education

Phd student, The University of Adelaide, Australia (supervised by Prof. Jian Zuo)

Adelaide

Focus on the impact of climate change on building energy, and carbon footprint of mass timber construction

01.2024-present

Master of Architecture, Tianjin University, China

Tianjin

Average Score 91.18/100 (top 5%)

09.2020-06.2023

Thesis: Life cycle assessment of mass timber construction in China

Bachelor of Architecture, Wuhan University of Technology, China

Wuhan, Hubei

Average Score 89.42/100 (top 5%)

09.2015-06.2020

Focus on post occupancy evaluation, campus space activation



[Google scholar](#)

Publications

Duan, Z., Omrany, H., & Zuo, J. (2025). Exploring embodied greenhouse gas (GHG) emissions of mass timber construction: A comparative study of life cycle assessment databases. *Resources, Conservation and Recycling*, 223, 108491. **(JCR Q1, IF=10.9)**  

Duan, Z., de Wilde, P., Attia, S., & Zuo, J. (2025). Challenges in predicting the impact of climate change on thermal building performance through simulation: A systematic review. *Applied Energy*, 382, 125331. **(JCR Q1, IF=11.0)**  

Duan, Z., Omrany, H., & Zuo, J. (2025). Impact of climate change on energy performance and energy conservation measures effectiveness in Australian office buildings. *Energy*, 319, 134956. **(JCR Q1, IF=9.4)**  

Duan, Z., de Wilde, P., Attia, S., & Zuo, J. (2024). Prospect of energy conservation measures (ECMs) in buildings subject to climate change: A systematic review. *Energy and Buildings*, 322, 114739. **(JCR Q1, IF=7.1)**  

Duan, Z., (2023). Impact of climate change on the life cycle greenhouse gas emissions of cross-laminated timber and reinforced concrete buildings in China. *Journal of Cleaner Production*, 395, 136446. **(JCR Q1, IF=10.0)**  

Duan, Z., Huang, Q., & Zhang, Q. (2022). Life cycle assessment of mass timber construction: A review. *Building and Environment*, 221, 109320. **(JCR Q1, IF=7.6)**  

Duan, Z., Huang, Q., Sun, Q., & Zhang, Q. (2022). Comparative life cycle assessment of a reinforced concrete residential building with equivalent cross laminated timber alternatives in China. *Journal of Building Engineering*, 62, 105357. **(JCR Q1, IF=7.4)**  

Sun, Q., Huang, Q., **Duan, Z.**, & Zhang, A. (2022). Recycling Potential Comparison of Mass Timber Constructions and Concrete Buildings: A Case Study in China. *Sustainability*, 14(10), 6174. **(JCR Q2, IF=3.3)**  

Tang, S., **Duan, Z.***, Chen, W., & Qiao, H. (2022). Research on post occupancy evaluation of Oze National Park in Japan based on online reviews. *Journal of Asian Architecture and Building Engineering*, 22(2), 602-619. **(JCR Q4, IF=1.6)**  

Research

IEA EBC Annex 89: Implementing Net-Zero Whole Life Carbon Buildings (July 2024 – Present)

Adelaide

Research Collaborator (Supervised by Prof. Jian Zuo)

07.2024-present

- Link my PhD research with IEA Annex 89 objectives, fostering collaboration with global researchers focused on achieving net-zero carbon emissions.

National Key Research and Development Program of China (No.2016YFC0700201): Goal and effect-oriented green building design principles and methodologies

Tianjin

Researcher (Supervised by Prof. Qiong Huang)

09.2020-01.2023

- Conducted systematic review on life cycle assessment of mass timber construction
- Developed quantitative evaluations about the life cycle greenhouse gas emissions and life cycle primary energy of three residential buildings in Tianjin, China (RC/CLT/Hybrid CLT buildings)
- Assessed the impact of climate change on the heating and cooling energy demand

Solar Decathlon China: Tianjin U+

Key Member (Supervised by Prof. Wei Yang)

Zhangjiakou

03.2021-08.2022

- Took the lead on the design to reduce thermal bridges in the building envelope, particularly at joints between prefabricated and customizable modules
- Waterproof and breathable membrane installation in the building envelope and Life cycle cost

National Natural Science Foundation of China (No. 51808418)

Wuhan, Hubei

Researcher (Supervised by Prof. Sony Tang)

06.2019-09.2020

- Explored the post occupancy evaluation of national parks based on online reviews, by conducting content analysis, semantic analysis and topics analysis through python

Independent Innovation Project of Wuhan University of Technology (No. 2018-TJ-B1-09): Study on the Activation of Negative Space in University Campus from the Perspective of Space Acupuncture

Wuhan, Hubei

03.2018-03.2019

Researcher (Supervised by Prof. Yi Zhang)

- Conducted field survey and identified a "campus enclave" as an isolated, inactive campus space.

Teaching assistant

School of Architecture and Civil Engineering, The University of Adelaide

CONMGNT 7046 S2 24: Construction Management Technology I

08.2024-11.2024

CONMGMT 3008 S1 25: Digital Design and Construction: Coordination

03.2025-06.2025

Award

Terence Williamson Prize in Architectural Research	2025
Hans-Jürgen and Marianne Ohff Research Grant	2025
Ian Wilson Liberal Foundation Research Supplementary Scholarship	2025
CLIVE E. BOYCE Research Scholarship in Architecture	2025
Research Training Program Scholarship	2024
First place in 3rd Solar Decathlon China (Key member)	2022
China National Scholarship (0.2%)	2022
Tianjin University first-class scholarship (Two times) (10%)	2021,2022
Triple-A student at Tianjin University	2021,2022
Tianjin University second-class scholarship (30%)	2020
Outstanding Graduates of Wuhan University of Technology (WUT)	2020
Outstanding student cadres (Three times at WUT)	2016-2019
National Encouragement Scholarship (Four times at WUT) (3%)	2015-2019

TECHNICAL & LANGUAGE SKILLS

Building 3D modeling: AutoCAD, Revit, SketchUp, Rhinoceros, Grasshopper,

Post-processing: Photoshop, Illustrator, InDesign

IELTS 7.0 (Listening 8.5, Reading 7.0, Writing 6.0, Speaking 6.0)

21.08.2022

Reviewer for Nature Cities, Building and Environment, Communications Earth & Environment,

Journal of Cleaner Production, Journal of Building Engineering, Intelligent Buildings International,

Sustainable Built Environment (SBE) Regional Conference 2025

Referees

Jian Zuo (Professor, School of Architecture and Civil Engineering, Adelaide University, Australia)	jian.zuo@adelaide.edu.au
Shady Attia (Professor, Sustainable Architecture & Building Technology, Liège University, Belgium)	Shady.Attia@uliege.be
Hossein Omrany (Lecture, Construction Management, Adelaide University, Australia)	hossein.omrany@adelaide.edu.au
Qiong Huang (Associate Professor, School of Architecture, Tianjin University, China)	qhuang@tju.edu.cn
Wei Yang (Professor, School of Architecture, Tianjin University, China)	walker_yang@tju.edu.cn